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Paper:

Traversability-Based RRT* for Planetary Rover Path Planning in Rough Terrain with LIDAR Point Cloud Data

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Sampling-based search algorithms such as Rapidly-Exploring Random Trees (RRT) have been utilized for mobile robot path planning and motion planning in high dimensional continuous spaces. This paper presents a path planning method for a planetary exploration rover in rough terrain. The proposed method exploits the framework of a sampling-based search, the optimal RRT (RRT*) algorithm. The terrain geometry used for planning is composed of point cloud data close to continuous space captured by a light detection and ranging (LIDAR) sensor. During the path planning phase, the proposed RRT* algorithm directly samples a point (node) from the LIDAR point cloud data. The path planner then considers the rough terrain traversability of the rover during the tree expansion process of RRT*. This process improves conventional RRT* in that the generated path is safe and feasible for the rover in rough terrain. In this paper, simulation study on the proposed path planning algorithm in various real terrain data confirms its usefulness.

Keywords: path planning, field robotics, RRT, planetary rover

1. Introduction

In extreme environments such as on Mars or in volcanic areas, mobile robots have been used in scientific missions or as precursors for future manned missions. Robot called planetary exploration rovers are typically operated from the Earth using remote control systems. However, the communication latency between the Earth and the rover often impedes the mission. Therefore, autonomous mobility systems have been essential for safe rover operation. These systems consist of three major processes. First, the rover detects the environment using cameras or other sensors and creates a terrain map of the surrounding area. After that process, the rover finds a traversable path and follows the path based on its navigation and control subsystems. These processes are managed by a space-qualified, radiation-hardened onboard computer. The processing power of such computers is often limited owing to limitations in the power supply. Thus, a computationally efficient algorithm for the autonomous system is favorable.

During the environmental perception phase, the Mars rover typically uses onboard cameras. However, such cameras often have a narrow field of view, and capturing high quality images requires sufficient ambient light. Some alternative systems, namely light detection and ranging (LIDAR) sensors and laser range finder (LRF) sensors have been exploited in autonomous cars [1, 2]. These sensors are yet to be exploited for rover missions. They can measure 3D terrain geometry and provide a wider field of view compared to traditional cameras. The 3D terrain geometry data, called point cloud data, include an enormous amount of data points. Therefore, they need to be processed efficiently for path planning from a computational burden perspective.

Path planning is an important process for robots to reach a goal safely in rough terrain. Two main types of path planning method have been used previously: grid-based and sampling-based methods. For planetary exploration rovers, the grid-based path planning method has been addressed in many research papers [3–5]. Most of these methods are extensions of Grid-based Estimation of Surface Traversability Applied to Local Terrain (GESTALT) [4]. This method first converts the terrain geometry into a grid-based digital elevation model (DEM). An optimal path is then generated based on a cost function composed of terrain information. Grid-based search algorithms are often used to find the path that provides the minimum value for total cost. However, the computations required for this approach often increase exponentially as the dimensionality of the problem, or the resolution of the grid increases. Additionally, a discretized map often fails to represent complicated features in rough terrain, such as ditches, irregular rocks and other obstacles. Some studies have attempted to solve this issue by modifying the grid shape or applying filters [5–8], but the path generated in different grids changes based on the resolution or shape of the grids.

As compared with grid-based methods, sampling-based methods are typically used in continuous spaces. Rapidly-Exploring Random Trees (RRT) have been widely used in sampling-based path planning for mobile robots [9]. This method is computationally efficient, particularly in high

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dimensional continuous spaces, although it does not find a completely optimal path. There have been many studies focusing on improvement of the optimality of RRT paths [10–12]. For example, RRT* is one of the algorithms focusing on path optimality. RRT* proposed by S. Karaman et al. [10] can asymptotically find an optimal path depending on the number of sampling trials. In the method, once a path is found from an initial trial, the method performs sampling again and evaluates if the new path is more optimal than the previous path. These works were performed in high dimensional continuous spaces with scattered objects. RRT has been also used in autonomous cars in real world environments [13]. This research was conducted on a paved roadway, which is a relatively smooth 2D terrain, although there were some dynamic objects, including pedestrians, cars and other obstacles. However, this method has not been applied to rough terrain. Belter et al. [14] proposed an RRT-based path planning method for a six-legged walking robot in rough terrain using a laser scanner and stereo camera. Although some have proposed sampling-based path planning methods that consider terrain information [15–18], a sampling-based path planning algorithm based on rover traversability in three dimensional rough terrain is an open issue.

This paper aims to develop a sampling-based path planning method for planetary exploration rovers in rough terrain. The basic algorithm for path planning employs RRT* because it can find paths in continuous space efficiently as the sampled paths asymptotically converge to the optimal path. The proposed method improves the tree expansion process of conventional RRT* by considering rover traversability, configuration, and the goal area. A node randomly sampled from the point cloud data is stored in the tree trunk once the node undergoes a traversability assessment. The traversability assessment is mainly based on a user-defined area and the travel cost between two nodes. The defined area for rover traversability and a goal bias restricts tree expansion. Here, the node is connected to a single parent node from the possible candidate nodes such that the cost between the sampled node and the parent node is minimized. The travel cost is based on the distance and the rover configuration (roll, pitch, and yaw angles). Additionally, this method directly uses point cloud data close to continuous spaces captured by LIDAR. This approach achieves computationally-efficient path planner.

Simulation studies using the proposed method on various types of fractal and real terrain data are presented later in this paper. The effectiveness of the traversability assessment is confirmed by varying the parameters used.

The rest of this paper is organized as follows: Section 2 describes the proposed method in detail; Section 3 discusses the simulations and evaluations; and finally, conclusions are summarized in Section 4.

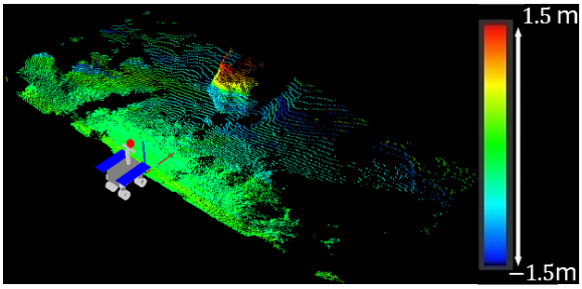


Fig. 1. An example of terrain features expressed by point cloud data.

2. Path Planning Method

The path planning method proposed in this paper consists of the following steps: a terrain map captured by LIDAR is provided as point cloud data; then a randomized node sampling for tree expansion is performed with the rover traversability assessment; finally, a traversable path and tree for the rover are achieved.

2.1. Terrain Mapping

This research assumes that LIDAR is used for terrain mapping. LIDAR measures terrain features, such as obstacle sizes, slope angles, and roughness as a set of point cloud data. **Fig. 1** presents an example of point cloud data describing rough terrain. This terrain data contains approximately 80,000 points and describes the area within a six-meter radius from the rover. As shown in **Fig. 1**, the density of the point cloud data varies depending on distance from the LIDAR scanner. The data points near the initial state (starting position for rover movement) are more densely sampled. In other words, the sampling algorithm is naturally biased toward the initial rover position. In this paper, each point is numbered and called a node. Furthermore, a line between nodes in a map is defined as an edge.

2.2. RRT* Algorithm

The basic RRT algorithm uses a randomized tree data structure to find a path by exploring a search space efficiently based on a random function [9]. This algorithm is useful for solving problems with a high degree of freedom and nonholonomic constraints owing to its high computational efficiency. However, RRT does not explicitly consider path-quality (i.e., how safe is a path for rover travel), nor does it have a process to improve path quality during tree expansion. Therefore, RRT does not guarantee path optimality. RRT* overcomes this problem by introducing incremental rerouting of the edges between pairs of nodes in the tree. Randomly sampled nodes are not only added to a tree, but also considered as replacements for the parents of other nodes in the tree. Additionally, RRT* selects parent nodes that minimize the cost between the start node and randomly chosen other nodes during tree expansion. As a results, RRT* asymptotically finds the optimal solution to the path planning problem.

Algorithm 1 Traversability-based RRT*.

Input: C-space: $\mathbb{C}^6$ at pointcloud data: Pc
Output: the tree: T

1: T.addvertex($\boldsymbol{q}_{init}$);
2: $cost(\boldsymbol{q}_{init}) = 0$
3: **while** $\boldsymbol{q}_i \neq \boldsymbol{q}_{goal}$ **do**
4: $\quad \boldsymbol{q}_i \leftarrow$GetRandomNode();
5: $\quad$ **for all** $\boldsymbol{q}_j \in T_{near}$ **do**
6: $\quad\quad$ **if** Traversability Assessment($\boldsymbol{q}_j, \boldsymbol{q}_i$) **then**
7: $\quad\quad\quad cost = cost(\boldsymbol{q}_j)$+CostFUNC($\boldsymbol{q}_j, \boldsymbol{q}_i$);
8: $\quad\quad\quad$ **if** $cost < cost(\boldsymbol{q}_i)$ **then**
9: $\quad\quad\quad\quad cost(\boldsymbol{q}_i) = cost$
10: $\quad\quad\quad\quad parent(\boldsymbol{q}_i) = \boldsymbol{q}_j$
11: $\quad\quad\quad\quad \boldsymbol{q}_n = \boldsymbol{q}_j$
12: $\quad T$.addvertex($\boldsymbol{q}_i$)
13: $\quad T$.addedge($\boldsymbol{q}_n, \boldsymbol{q}_i$)
14: $\quad$ **for all** $\boldsymbol{q}_{near} \in T_{near}$ except $\boldsymbol{q}_i$ **do**
15: $\quad\quad$ **if** Traversability Assessment($\boldsymbol{q}_i, \boldsymbol{q}_{near}$) **then**
16: $\quad\quad\quad cost = cost(\boldsymbol{q}_i)$+COSTFUNC($\boldsymbol{q}_i, \boldsymbol{q}_{near}$);
17: $\quad\quad\quad$ **if** $cost(\boldsymbol{q}_{near}) > cost$ **then**
18: $\quad\quad\quad\quad T$.rewire($\boldsymbol{q}_i, \boldsymbol{q}_{near}$);
19: $\quad\quad\quad\quad cost(\boldsymbol{q}_{near}) = cost$
20: $\quad\quad\quad\quad parent(\boldsymbol{q}_{near}) = \boldsymbol{q}_i$
21: **return** T

2.3. Traversability-Based RRT*

The RRT* algorithm typically samples a node from a huge space within the point cloud data. Such sampling may extend the tree between a parent node and the randomly selected node, even though the rover cannot physically traverse the tree (or edge). To solve this issue, a traversability assessment is introduced as a user-defined area derived from rover traversability on the terrain. Tree expansion is generally preferable when it spreads toward a goal through a relatively level area. Therefore, in our approach, the tree expansion area is limited based on the rover dimensions, configuration, and a heading relative to the goal.

Algorithm 1 presents the procedure for the proposed method, called traversability-based RRT*. Initially, the proposed method operates similarly to basic RRT* (**Algorithm 1**, Lines 1-3). Here, $\boldsymbol{q}_i$ is the rover configuration given by:

$$\boldsymbol{q}_i = (x_i, y_i, z_i, \phi_i, \theta_i, \psi_i) \in Pc \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (1)$$

where, i is an index of a node or point cloud. x_i, y_i, and z_i are the coordinates of the center of gravity of the rover. ϕ_i, θ_i, and ψ_i are the posture angle, roll, pitch, and yaw of the rover, respectively. The edge between a randomly selected configuration ($\boldsymbol{q}_i$) and one of the configurations ($\boldsymbol{q}_j$) in the tree near $\boldsymbol{q}_i$ is checked using a traversability assessment (**Algorithm 1**, Line 5). If the edge passes the traversability assessment, the cost from the start configuration to $\boldsymbol{q}_i$ via $\boldsymbol{q}_j$ is calculated by using a cost function (**Algorithm 1**, Line 6). The cost function is defined as follows:

$$C(\boldsymbol{q}_i) = W_L \frac{L_i}{N_L} + W_\phi \frac{\phi_i}{N_\phi} + W_\theta \frac{\theta_i}{N_\theta} + W_\psi \frac{\Delta\psi_i}{N_\psi} \quad . \quad (2)$$

where $C(\boldsymbol{q}_i)$ is the cost from $\boldsymbol{q}_j$ to $\boldsymbol{q}_i$. L_i is the distance between node i and node j. ϕ_i and θ_i are the roll and pitch angles of the rover at node i respectively. $\Delta\psi_i$ is the rotation yaw angle from node j to node i. W_L, W_ϕ, W_θ, and W_ψ are the weighting factors for each index. N_L, N_ϕ, N_θ, and N_ψ are the normalization factors to make each index dimensionless. Here, the parent configuration $\boldsymbol{q}_n$ is selected using Eq. (3) (**Algorithm 1**, Line 7-11) and $\boldsymbol{q}_i$ is stored in the tree as a new node (**Algorithm 1**, Line 12-13).

$$\boldsymbol{q}_n = \underset{\boldsymbol{q}_j \in T_{near}}{\arg\min}(cost(\boldsymbol{q}_j) + C(\boldsymbol{q}_i)) \quad . \quad . \quad . \quad . \quad . \quad . \quad (3)$$

$cost(\boldsymbol{q}_j)$ is defined as follows:

$$cost(\boldsymbol{q}_j) = \sum_{k \in \boldsymbol{wp}_j} C(\boldsymbol{q}_k). \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (4)$$

$\boldsymbol{wp}_j$ consists of a series of chained nodes, which form the set of waypoints from the start node to node j:

$$\boldsymbol{wp}_j = (s, \ldots, j), \ \ \boldsymbol{q}_s = \boldsymbol{q}_{init}. \quad . \quad . \quad . \quad . \quad . \quad . \quad (5)$$

To improve optimality, RRT* is applied in the tree rewiring phase (**Algorithm 1**, Line 14-20). In this phase, the cost of $\boldsymbol{q}_{near}$ to $\boldsymbol{q}_i$ is compared with a new cost, derived from $\boldsymbol{q}_i$ to $\boldsymbol{q}_{near}$. If the new cost is less than the previous cost $cost(\boldsymbol{q}_{near})$, then the tree is rewired. The edge between $parent(\boldsymbol{q}_{near})$ and $\boldsymbol{q}_{near}$ is erased, and a new edge between $\boldsymbol{q}_i$ and $\boldsymbol{q}_{near}$ is generated. This process can generate costless edges based on the sampling trials. Lines 3 to 20 of **Algorithm 1** are repeated until the generated tree reaches near the goal state. Using this process, the traversability-based RRT* algorithm provides a feasible/traversable and suboptimal path for the rover. In next subsection, the traversability assessment is explained in detail.

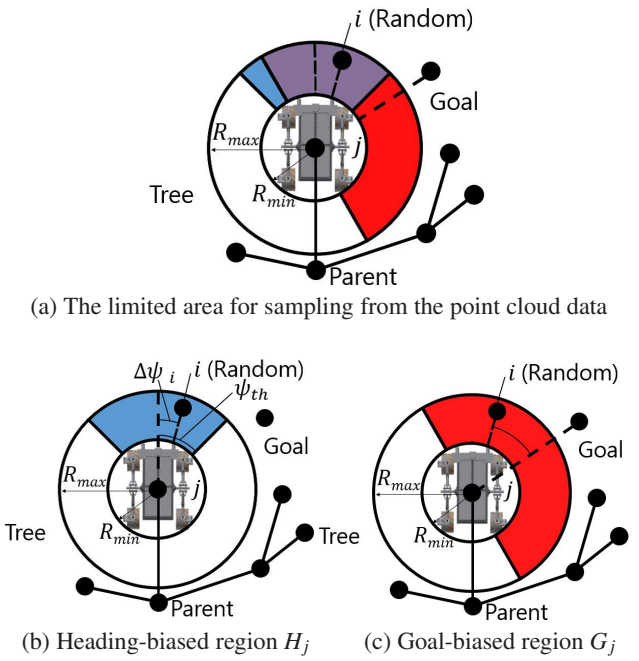
Algorithm 2 Traversability assessment.

Input: $\boldsymbol{q}_j, \boldsymbol{q}_i$
Output: True or False

if $\boldsymbol{q}_i \in H_j$ and $\boldsymbol{q}_i \in G_j$ and $|\delta_i| < |\delta_{th}|$ **then**
$\quad \phi_i, \theta_i \leftarrow$CALCPOS($\boldsymbol{q}_i$)
$\quad$ **if** $\phi_i < \phi_{th}$ and $\theta_i < \theta_{th}$ **then**
$\quad\quad$ **return** True
$\quad$ **else**
$\quad\quad$ **return** False
else
$\quad$ **return** False

2.4. Traversability Assessment

Traversability assessment is performed in lines 6 and 15 of **Algorithm 1**. **Algorithm 2** shows the procedure for this assessment. Initially, the traversability assessment checks if $\boldsymbol{q}_i$ is in the areas H_j and G_j (**Algorithm 2**,



(a) The limited area for sampling from the point cloud data

(b) Heading-biased region H_j (c) Goal-biased region G_j

Fig. 2. The limited area for sampling from the point cloud data.

Line 1). As shown in **Fig. 2(a)**, assuming that the rover turns toward node j at the *parent* node and tends to move straight forward, the tree expansion area is limited to the overlapping area. This area is composed of H_j and G_j each of which are defined by both radial and angular coordinates as follows: the radial distance from node j to node i is limited to the range between R_{min} and R_{max}. In **Fig. 2(b)**, for the angular direction, assuming that the current rover heading agrees with the direction from node *parent* to node j, the "field of view region" (heading-biased region : H_j) for the rover is determined by applying a threshold angle of $\pm\psi_{th}$ from the rover facing direction at node j. Subsequently, a goal-biased region (G_j in **Fig. 2(c)**) is also determined, the range of which is $\pm\pi/2$ measured from the direction between node j and the goal. Finally, the tree expansion area is obtained as the intersection of these two regions, as shown in **Fig. 2(a)**. Furthermore, the elevation angle between node j and node i is calculated as follows:

$$\delta_i = \arctan \frac{(z_i - z_j)}{\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}}. \quad . \; . \; . \quad (6)$$

If the elevation angle is less than the threshold elevation angle δ_{th}, the traversability assessment steps forward. The roll and pitch angles of the rover are calculated at this point (**Algorithm 2**, Line 2). This calculation leverages the terrain geometry. As shown in **Fig. 3**, each wheel coordinate for the rover is given as follows:

$$\begin{bmatrix} x_m \\ y_m \end{bmatrix} = \frac{l}{2} \begin{bmatrix} \cos\psi_i & -\sin\psi_i \\ \sin\psi_i & \cos\psi_i \end{bmatrix} \begin{bmatrix} \sigma(x_m)\sin(\alpha) \\ \sigma(y_m)\cos(\alpha) \end{bmatrix} + \boldsymbol{x}_{Gi} \quad (7)$$

where, l is the distance between the left front and right rear wheels, and m is the index of each wheel. α is the absolute angle for the ratio of the rover's tread to its wheel-

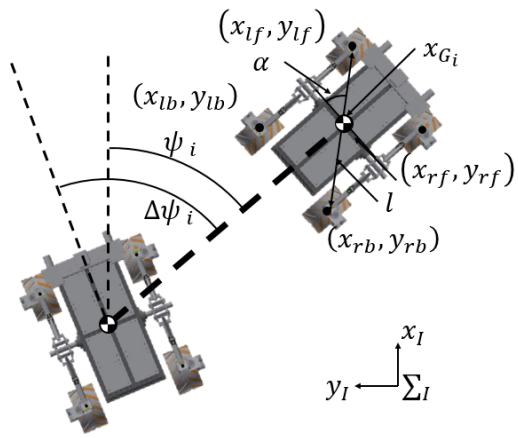


Fig. 3. The wheels coordinates.

base, and $\boldsymbol{x}_{Gi}$ is the rover's center of gravity at node i. $\sigma(x_m)$ and $\sigma(y_m)$ are defined as follows:

$$\begin{bmatrix} \sigma(x_{lf}) & \sigma(x_{lb}) & \sigma(x_{rf}) & \sigma(x_{rb}) \\ \sigma(y_{lf}) & \sigma(y_{lb}) & \sigma(y_{rf}) & \sigma(y_{rb}) \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix}. \quad (8)$$

The height coordinate of each wheel (z_m) is derived by calculating the average of the point cloud data surrounding each wheel contact point. The point cloud data used for the calculation lies within the area determined by the wheel thickness. The roll and pitch angles are then obtained geometrically as the angle between the inertial coordinate and a pseudo-plane (P), composed of the wheel contact points in the projected region of the rover, as illustrated in **Figs. 4** and **5**. In this paper, the suspension mechanism of the rover is excluded for reduction of the computational burden. Four different terrain inclinations can be calculated from each subset of three wheel contact points. The roll and pitch angles used here are assumed to be the largest inclinations between these values:

$$\phi_i = \max_{m \le 4}[\phi(P_m)], \quad . \; . \; . \; . \; . \; . \; . \; . \; . \; . \; . \quad (9)$$

$$\theta_i = \max_{m \le 4}[\theta(P_m)]. \quad . \; . \; . \; . \; . \; . \; . \; . \; . \; . \; . \quad (10)$$

If these angles are less than the threshold angles (**Algorithm 2**, Line 3), then the traversability assessment returns true (**Algorithm 2**, Line 4). This indicates that the edge between $\boldsymbol{q}_i$ and $\boldsymbol{q}_j$ is traversable. By satisfying this limitation for tree expansion area, the rover can avoid steep and obstructed areas, as well as the areas with no point cloud data, while being biased toward the goal.

2.5. Evaluation Index

This subsection introduces an evaluation index for evaluating the path from the perspective of total cost, length, and rover configuration. The evaluation index for the path is defined by the following equation:

$$E(p) = cost(\boldsymbol{q}_g) = \sum_{k \in \boldsymbol{wp}_g} C(\boldsymbol{q}_k) \quad . \; . \; . \; . \; . \; . \quad (11)$$

where $E(p)$ is the total cost of the path from the initial state to goal state, and $\boldsymbol{q}_g$ is the goal state. Additionally,

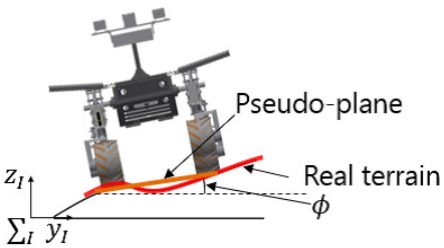


Fig. 4. Pseudo-plane and ϕ of the rover.

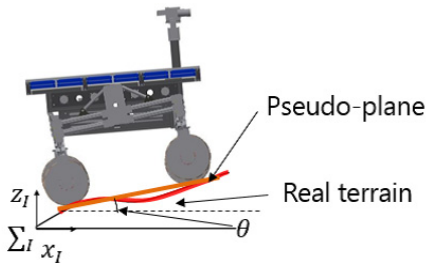


Fig. 5. Pseudo-plane and θ of the rover.

the evaluation indices for the length, as well as roll, pitch, and yaw angles are as follows:

$$E_L(p) = \sum_{k \in \boldsymbol{wp}_g} W_L \frac{L_k}{N_L} \quad \ldots\ldots\ldots\ldots (12)$$

$$E_\phi(p) = \sum_{k \in \boldsymbol{wp}_g} W_\phi \frac{\phi_i}{N_\phi} \quad \ldots\ldots\ldots\ldots (13)$$

$$E_\theta(p) = \sum_{k \in \boldsymbol{wp}_g} W_\theta \frac{\theta_i}{N_\theta} \quad \ldots\ldots\ldots\ldots (14)$$

$$E_\psi(p) = \sum_{k \in \boldsymbol{wp}_g} W_\psi \frac{\Delta\psi_i}{N_\psi}. \quad \ldots\ldots\ldots\ldots (15)$$

These evaluation indices can determine what aspect of the solution accounts for the largest part of $E(p)$.

3. Simulation Results and Discussion

3.1. Simulation Parameters

To confirm the usefulness of the proposed method, a path planning simulation was conducted on different types of rough real-world terrain, as well as a simulation map. The terrain map captured by LIDAR (UTM-30 LX-EW, developed by Hokuyo Corp.) in a volcanic area at Mt. Mihara in Japan is given as $[0.0, 6.0]$ meters and $[-90.0°, 90.0°]$. In this simulation, the rover parameters are based on a four-wheeled rover test bed developed by the author group. **Table 1** summarizes the parameters used for the simulation. R_{min}, R_{max}, ϕ_{th}, θ_{th}, and ψ_{th} are derived by determining the mobility of the rover test bed. Assuming that the rover can safely traverse an obstacle with a height of 150 mm, δ_{th} is set as 10.2°. The adaptability of the proposed method for various types of terrain and robots is verified by changing the parameters.

Table 1. Basic parameters for the proposed method.

Pramerter	Value
R_{max}	0.830 m
R_{min}	0.415 m
ϕ_{th}	10.0°
θ_{th}	10.0°
ψ_{th}	50.0°
δ_{th}	10.2°
$W_L, W_\phi, W_\theta, W_\psi$	0.25, 0.25, 0.25, 0.25
N_L	R_{max}
N_ϕ	ϕ_{th}
N_θ	θ_{th}
N_ψ	ψ_{th}

The following simulation scenarios were executed:

- Scenario 1: Simple path planning in a simulated terrain map.
- Scenario 2: Simple path planning in real terrain.
- Scenario 3: Sensitivity analysis of ψ_{th}.
- Scenario 4: Testing different values for weighting factors.

In Scenario 1, the goal was set at 12 m from the rover start position. In Scenarios 2, 3, and 4, the goal was randomly placed between 5 to 6 m from the start position. The results of each simulation are summarized in the following section.

3.2. Path Planning Simulation

In this subsection, the proposed method is tested on 25 different terrain data sets. Typical scenarios will be examined carefully in the following sections. Here, path planning is terminated once a node in the tree lies close to the goal, within a range of 0.3 m.

3.2.1. Path Planning in Varied Terrain and Sub-Optimal Paths

This simulation not only confirms the utility of the proposed method, but also attempts to find a sub-optimal path within a limited number of sampling trials. Here, node sampling continues even after the proposed method finds a path, allowing another path with less total cost when compared to the previously generated path to be found. By repeating this sampling, several paths can be found and the path between them with the lowest cost selected. The number of node sampling trials should be limited. In the scenarios described below, the number of node sampling trials is limited to 95,000 and 60,000, respectively, which is approximately the same number as the total points contained in the point cloud data for the terrain features. The typical results for path and tree expansion in Scenarios 1 and 2 are shown in **Figs. 6–8**. The trees and the path are depicted in **Fig. 6** and **Figs. 7–8**, respectively.

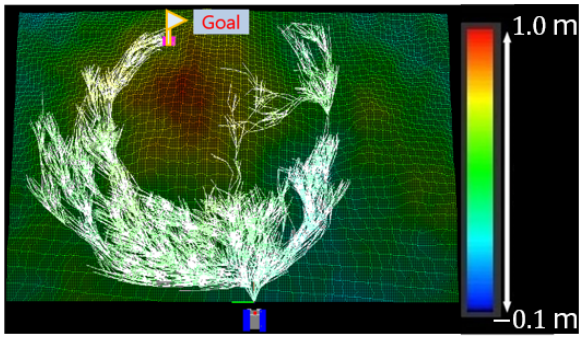


Fig. 6. An example of the tree in Scenario 1.

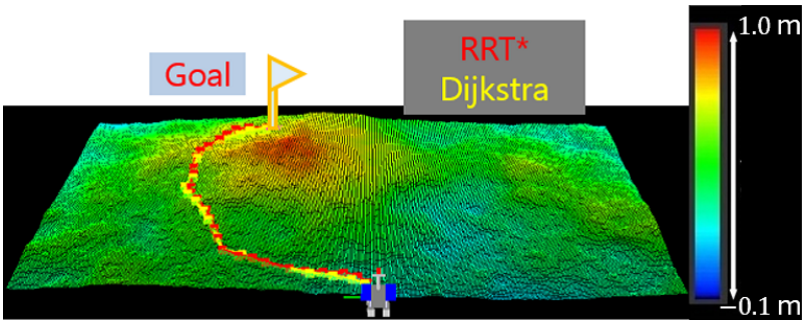


Fig. 7. An example of the paths in Scenario 1.

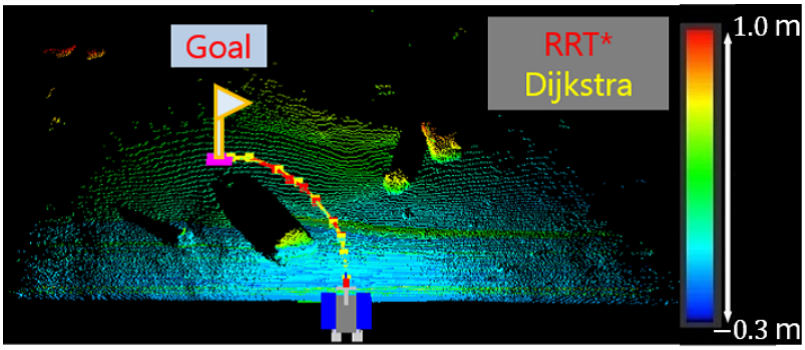


Fig. 8. An example of the paths in Scenario 2.

In Scenario 1, the proposed method could extend the tree to avoid hazardous areas, which are shown in the center of **Fig. 6**. Additionally, the minimum path cost is shown in **Fig. 7**. In Scenario 2, the generated path can reach the goal while avoiding obstacles in front of the rover as well and the occlusion area with no point cloud data (**Fig. 8**).

These results confirm that the proposed traversability based RRT* performed well in both real rough terrain and simulated terrain. These results also indicate that the rover can traverse areas filled with trees. In other words, the proposed path planning method simultaneously finds a single cost-minimum path and the traversable regions in the terrain.

As illustrated in **Figs. 7** and **8**, graph-based search method, such as the Dijkstra method, can also find optimal paths, although they require significant additional computational effort. For both scenarios, the optimal paths are generated Dijkstra algorithm in **Figs. 7** and **8**. The optimal path and the path generated by the proposed method are nearly identical. Therefore, it can be said that the proposed method generates a sub-optimal path with less computational effort.

Regarding the total cost of the paths $E(p)$, shown in **Figs. 9** and **10**, it is clear that the total cost decreases as the number of sampling trials increases. This trend does not depend on terrain features and the total cost converges at approximately 30,000 sampling trials. These figures demonstrate that traversability-based RRT* asymptotically converges faster than conventional RRT with a traversability assessment, and that the error rates between the convergence values of RRT* and the optimal path values are only 9% and 0.4%, respectively. These results confirm the utility of traversability-based RRT* both

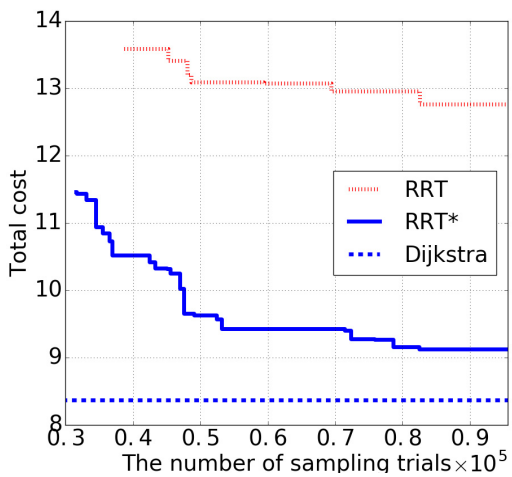


Fig. 9. Sampling trials and $E(p)$ on Scenario 1.

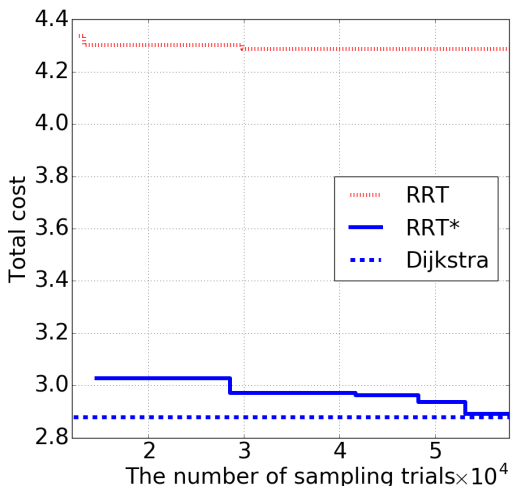


Fig. 10. Sampling trials and $E(p)$ on Scenario 2.

quantitatively and qualitatively when compared to the optimal path.

Regarding the difference between the two error rates, terrain size is the primary factor. In Scenario 1, there are many edges with an expectation for improvement due to the large map size. Thus, the error rate is larger than the rate in Scenario 2.

3.2.2. Sensitivity Analysis of ψ_{th}

Scenario 3 analyzes the sensitivity of the values of ψ_{th}. In the simulation, the value of ψ_{th} is set to 25°, 50°, or 75°. It should be noted that a larger ψ_{th} allows a wider area for sampling, as shown in **Fig. 2**. Simulations with each threshold value are performed in the rough terrain shown in **Fig. 11**.

The results of the simulations are summarized in **Figs. 11** and **12**. As seen in **Fig. 11**, there is very little difference between the three paths. However, it can be clearly seen that the computational time required to find a path increases as ψ_{th} increases, even when using the same number of sampling trials (**Fig. 12**). This is because

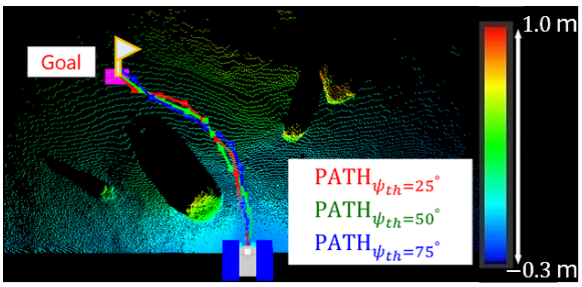


Fig. 11. An example of the tree and paths in Scenario 3.

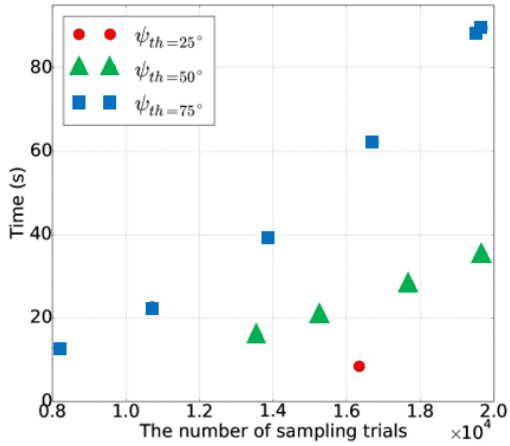


Fig. 12. Relationship between time and sampling trials.

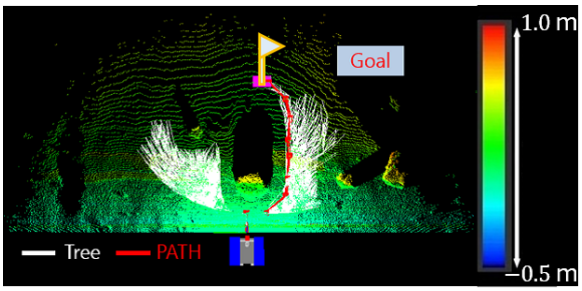


Fig. 13. An example path with $\psi_{th} = 90°$.

the tree expansion area is nearly unlimited with a large ψ_{th}. Therefore, path planning requires significant computational effort due to the large number of nodes in the tree. These tendencies can also be observed in the other simulation results.

On the other hand, if we use an excessively low ψ_{th}, then a path cannot be generated in cases where obstacles or ditches exist in front of the rover. As shown in **Fig. 13**, while the path cannot be obtained by using small ψ_{th}, a large ψ_{th}, such as 90° relaxes the sampling area and the path planner successfully finds path.

In summary, the heading threshold is a user-defined parameter, the value of which should be appropriately tuned based on terrain characteristics (i.e., obstacle-rich terrain, or regular but sloped terrain) and rover maneuverability (i.e., how easily the rover steers). This point remains a future work of this research.

Table 2. Weighting factors for traversability-based RRT*.

Path	W_L	W_ϕ	W_θ	W_ψ
Path A	0.25	0.25	0.25	0.25
Path B	0.40	0.10	0.10	0.40
Path C	0.10	0.40	0.40	0.10

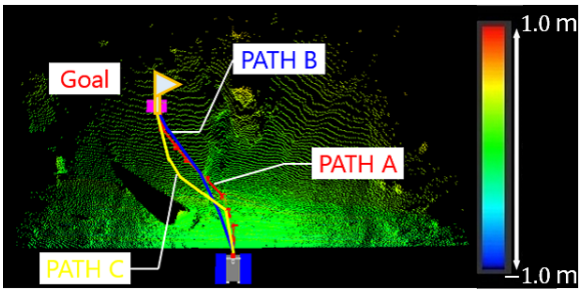


Fig. 14. An example of the paths in Scenario 4.

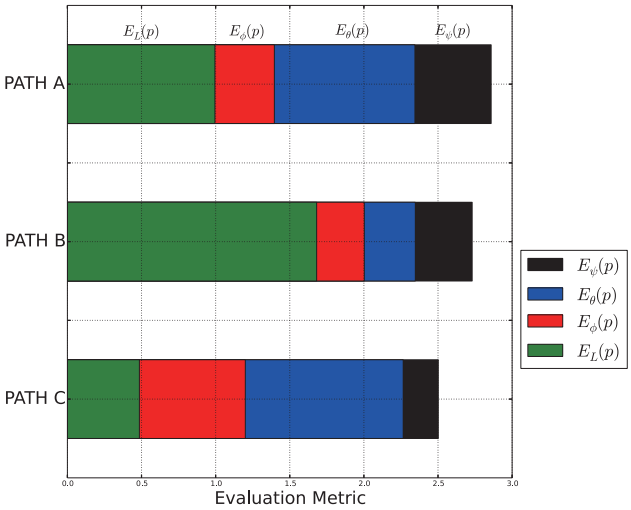


Fig. 15. Costs of path components in Scenario 4.

3.2.3. Different Values for Weighting Factors

Traversability-based RRT* uses a cost function that employs weighting factors (Eq. (2)). Scenario 4 examines the dependency effect of each weighting factor on the path planning result. Here, three combinations of the weighing factors (**Table 2**) are tested: PATH A considers each index equally, PATH B weights the distance and yaw angle, and PATH C weights the roll and pitch angles.

The simulations are performed 100 times under each condition. **Figs. 14** and **15**, and **Table 3** summarize the results. As shown in **Fig. 14**, there is a slight difference between the three paths. **Fig. 15** illustrates each evaluation index (Eqs. (12)–(15)) and indicates that the evaluation metric $E(p)$ mainly consists of $E_L(p)$ for PATH B, while $E_\phi(p)$ and $E_\theta(p)$ comprise the metric for PATH C. In **Table 3**, $\bar{\phi}$, $\bar{\theta}$, and $\Delta\bar{\psi}$ are defined as the average values throughout the path. Each evaluation index is generally

Table 3. The results of changing weighting factors in Scenario 4.

Path [-]	Total Distance [m]	$\bar{\phi}$ [°]	$\bar{\theta}$ [°]	$\Delta\bar{\psi}$ [°]
Path A	4.81	2.45	3.91	8.25
Path B	4.85	3.33	4.59	7.86
Path C	4.97	1.94	3.70	12.9

the lowest when the corresponding weighting factor takes on a value of 0.4. In these cases, the tree extends if the terrain inclination is modest and the tree shrinks to within a limited range of the rover heading if the terrain is steep.

This result confirms the effects of each weighting factor on path planning. Each weighting factor should be assigned based on user-demands (i.e., how short is path the rover traverses or how flat is the terrain?).

The total distance of PATH B is not the shortest when W_L is 0.4, but W_ψ is also 0.4. This may be because W_L does not influence path length directly in this type of terrain. This is because the rover heading is weighted such that the nodes can be arbitrarily sampled from a narrower area when compared to the case of a low W_ψ value. The trees then tend not to extend over wider areas where the shortest path may exist.

This result implies that W_ψ affects $\Delta\bar{\psi}$ as well as the total distance of the path, and is more effective than W_L for finding a shorter path; however, W_L is still useful in cases where a shorter path is preferable while still allowing for a large detour (i.e., avoiding large obstacles).

4. Conclusion and Future Work

In this paper, the traversability-based RRT* path planning method has been proposed for an application to the planetary exploration rovers in rough terrain.

The cost function and tree expansion limitation for the rover traversability assessment performed during the planning process ensure that the path is generated only in safely traversable regions of a terrain map.

Four different simulation studies using the proposed method were performed. The simulation results confirmed the usefulness of the proposed method in real rough terrain through a comparison with Dijkstra algorithm. The sensitivity of the threshold values used for the algorithm and the effects of the weighting factors have been also discussed: these values can be tuned based on terrain features and rover maneuverability. Auto-tuning of these values for practical exploration missions will be considered in future work.

Other future work includes considering wheel slippage on rough terrain. The surfaces of the Moon and Mars are covered with loose soil. Therefore, the wheels of the rover often slip or get stuck [4]. A possible future direction for this study is to incorporate the slip ratio into the cost function.

RRT* can find paths in high dimensional problems. Therefore, RRT* can effectively guide the motion planning for a rover in rough terrain while addressing time constraints, wheel velocity, steering maneuvers, and power efficiency.

The optimality of the path generated by the proposed method can be evaluated through a comparison with the path generated by a graph-search based algorithm with respect to computational efficiency. This provides a comparative evaluation of the path optimality and computational effort for each path planning procedure.

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